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INTEL DEVELOPER FORUM

Software Optimizations Become Simple with Top-Down Analysis Methodology on Intel® Microarchitecture, Code Name Skylake

Ahmad Yasin, Senior CPU Architect, Intel Corporation

ARCS002

intel
experience
what's inside™

Agenda

- Performance Analysis & Challenges
- Top-down Microarchitecture Analysis Method
- Demo using Intel® VTune™ Amplifier XE
- Skylake Performance Monitoring Unit Enhancements
- Summary and Q/A

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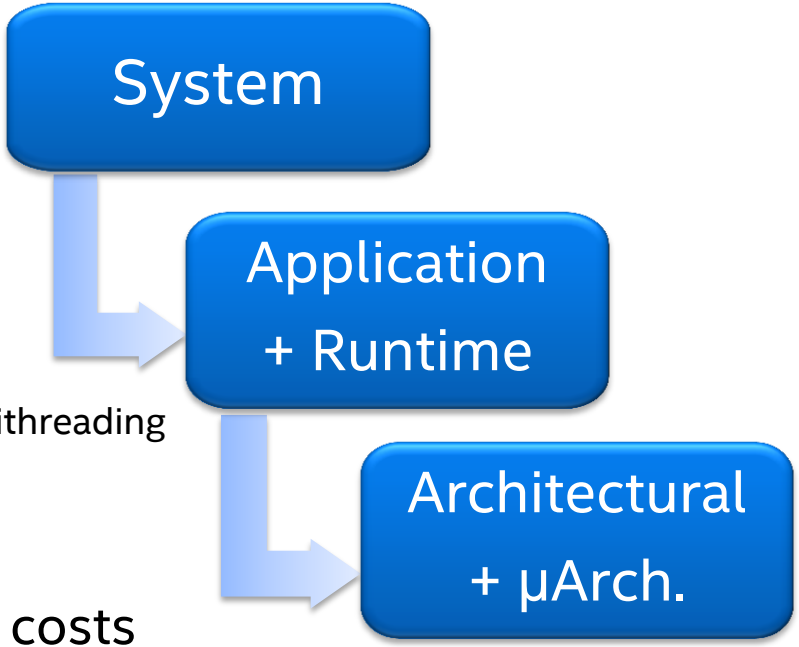
Background

- Performance Analysis Process
  - System Level
    - Examples: Disk, Network, Graphics
  - Application Level
    - Examples: Algorithm complexity, Parallelization/Multithreading
  - Architectural and microarchitectural Levels
    - Examples: vectorize key loops, reduce cache misses
- **Goal: detect bottlenecks** & estimate their costs
  - 2nd goal: locate issues (event-based sampling)
  - Not-a-goal: quantify speedup
  - Assumption: CPU Bound workload



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Preface

- Problem: Performance Optimization Is Difficult
  - Sophisticated microarchitectures
  - Application/workload diversity
  - Unmanageable profiling data
  - Tougher constraints (Time To Market, Resources, Priorities)
- Solution: **T**op-down **M**icroarchitecture **A**nalysis **M**ethod (TMAM)
  - Identifies **true** μ-arch bottlenecks in a simple, **structured** hierarchical process
  - **Analysis is made easier** for users who may lack hardware expertise
    - Simplified hierarchy avoids the μ-arch high-learning curve
    - Learn method once; use it on all Intel Architecture (IA) based processors
      - TMAM full hierarchy is supported on Core™ and Xeon™ processors
      - TMAM level-1 is available with recent generations of Atom™ and Xeon Phi™ processors

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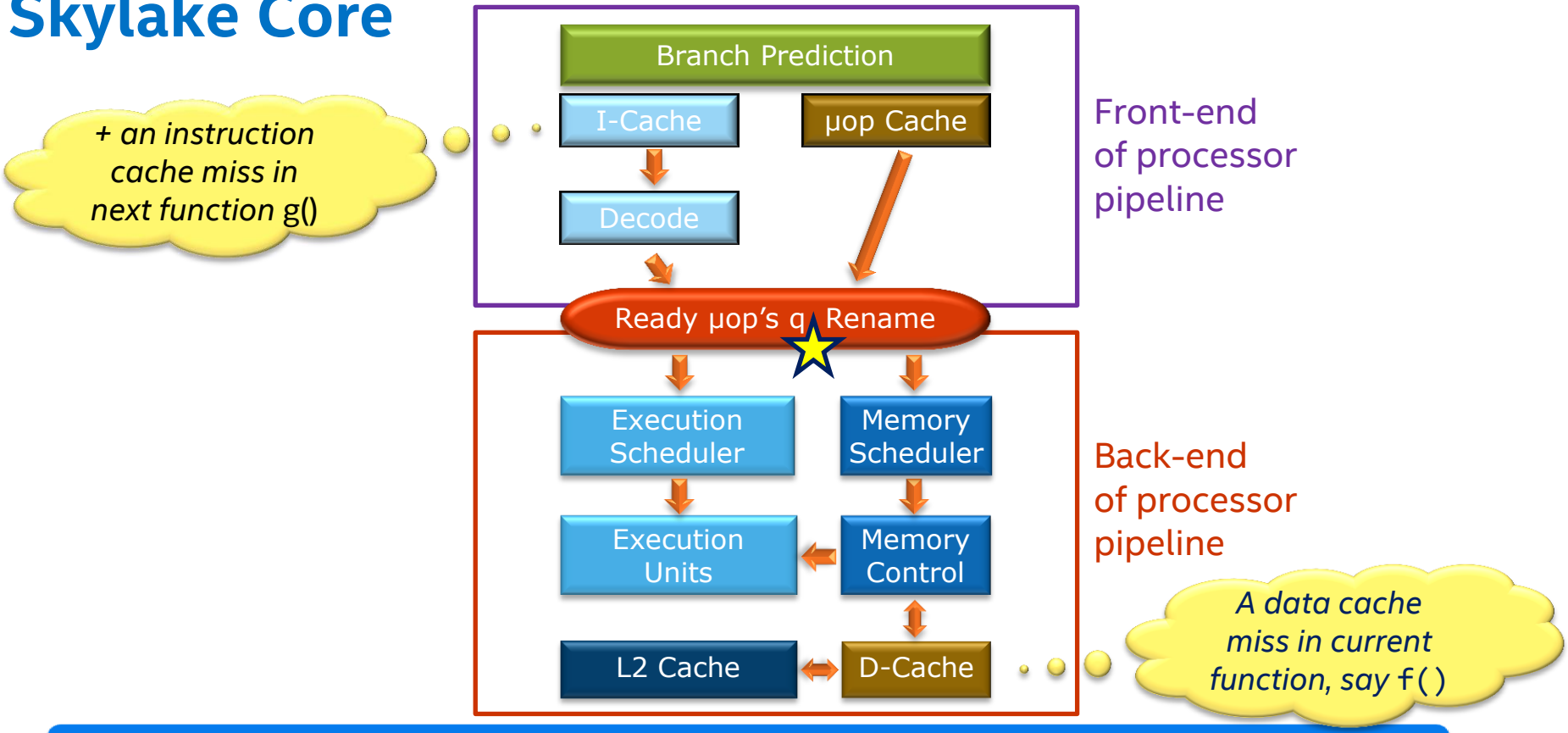
Skylake Core



TMAM is designed to help developer to focus on areas that matter

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Challenges

Traditional Methods

- Naïve approach

$Stall_Cycles = \Sigma\ Penalty_i * MissEvent_i$

- Unsuitable for modern out-of-order cores due to (Gaps):
  1) *Stalls Overlap*
  2) Speculative Execution
  3) Workload-dependent penalties
  4) Predefined set of miss-events
  5) Superscalar inaccuracy

Top Down Analysis

- A hierarchy
  - Top-down designated events at appropriate pipeline stages
  - "Hierarchical safety property"
- Addressing Gaps
  - **Bad Speculation at the top**
  - Generic top-down orients events that
    - count when matters; e.g. stalls vs cycles
    - count where matters; e.g. single point of division
    - count at finer-grain; e.g. issue-pipeline width.

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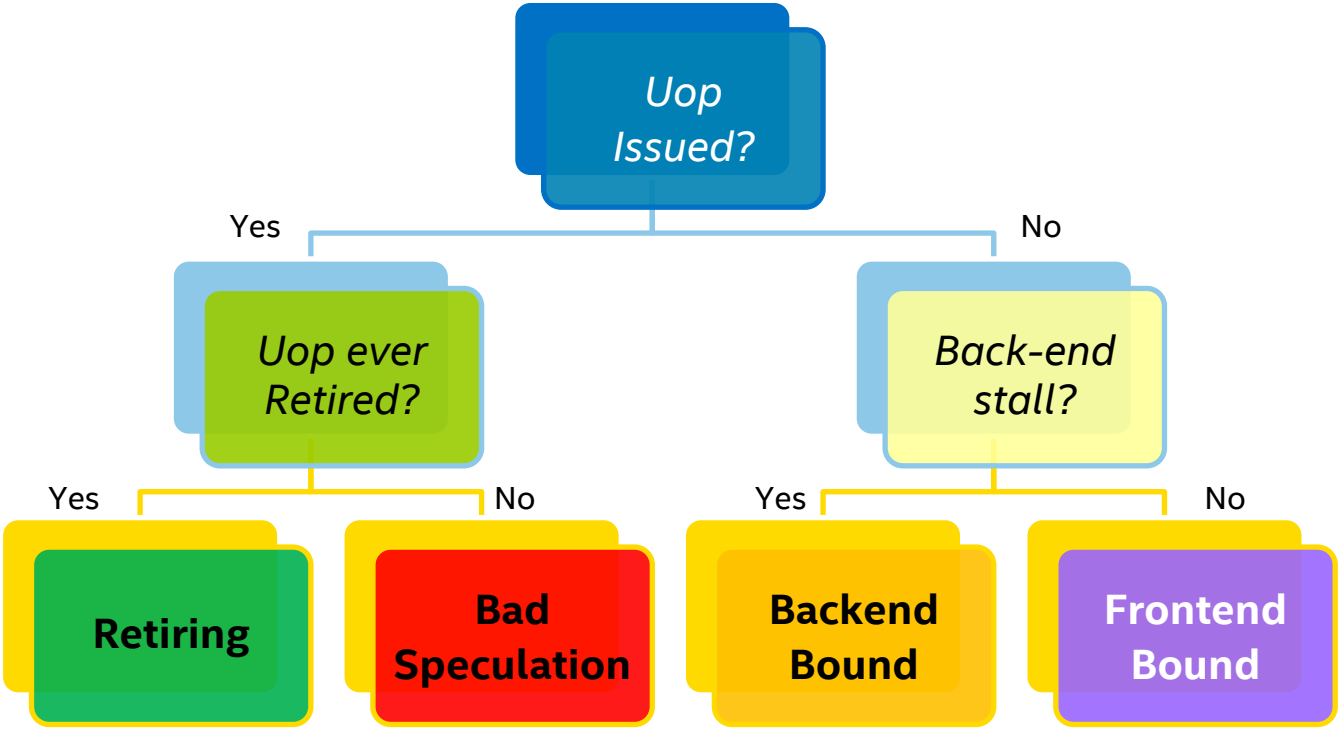
Top Level Breakdown



Uop := micro-operation. Each x86 instruction is decoded into uop(s)
Uop Issue := last front-end stage where a uop is ready to acquire back-end resources
Back-end stall := Any backend resource fills up which blocks issue of new uops

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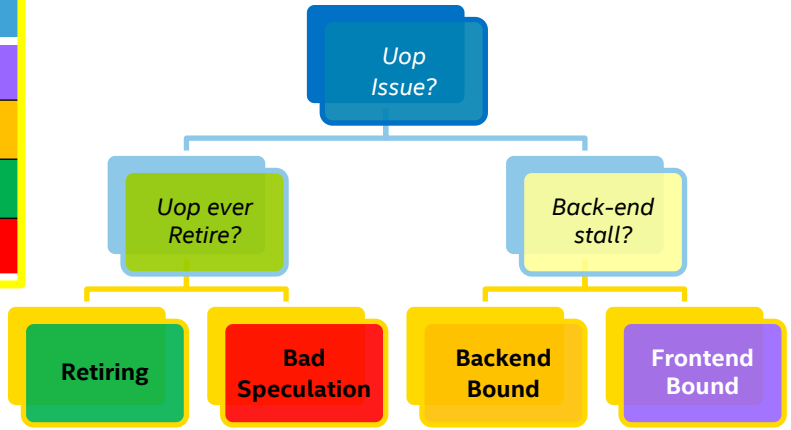
Top Level Breakdown Illustration

Cycle	1	2	3	4	…	x
Back-end Stall	0	0	1	0		0
Issue Slot 0	-	v	-	v		v
Issue Slot 1	-	v	-	v		v
Issue Slot 2	-	-	-	v		v
Issue Slot 3	-	-	-	v		-
Frontend Bound	4	2		0		1
Backend Bound			4	0		0
Retiring		2		1		2
Bad Speculation				3		1

Issue Slots := number of uops the front-end pipeline can issue to the back-end in a cycle.

Bubble := An un-utilized Slot.



Classify Each Issue Slot Into 1 of 4 Categories

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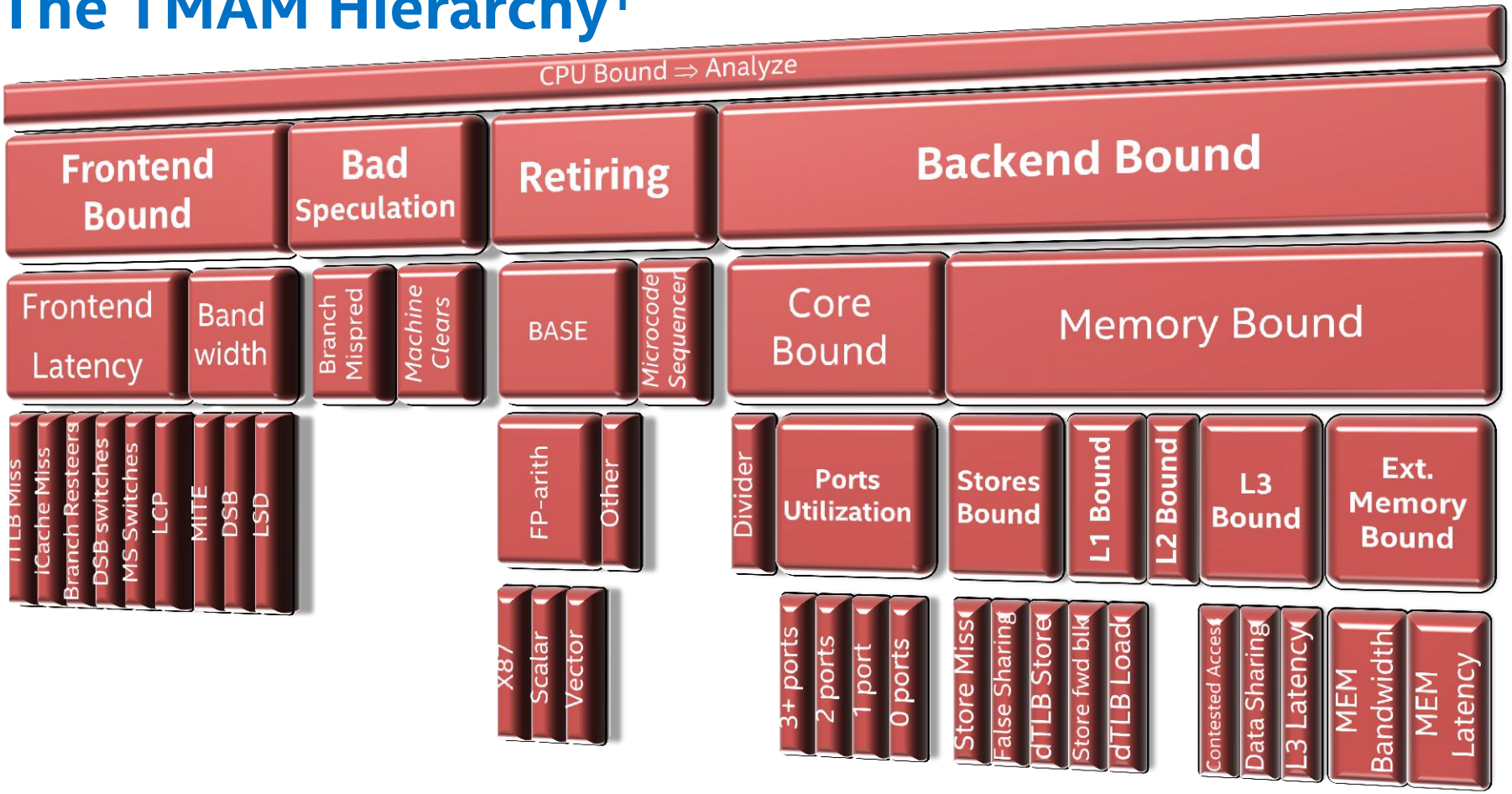
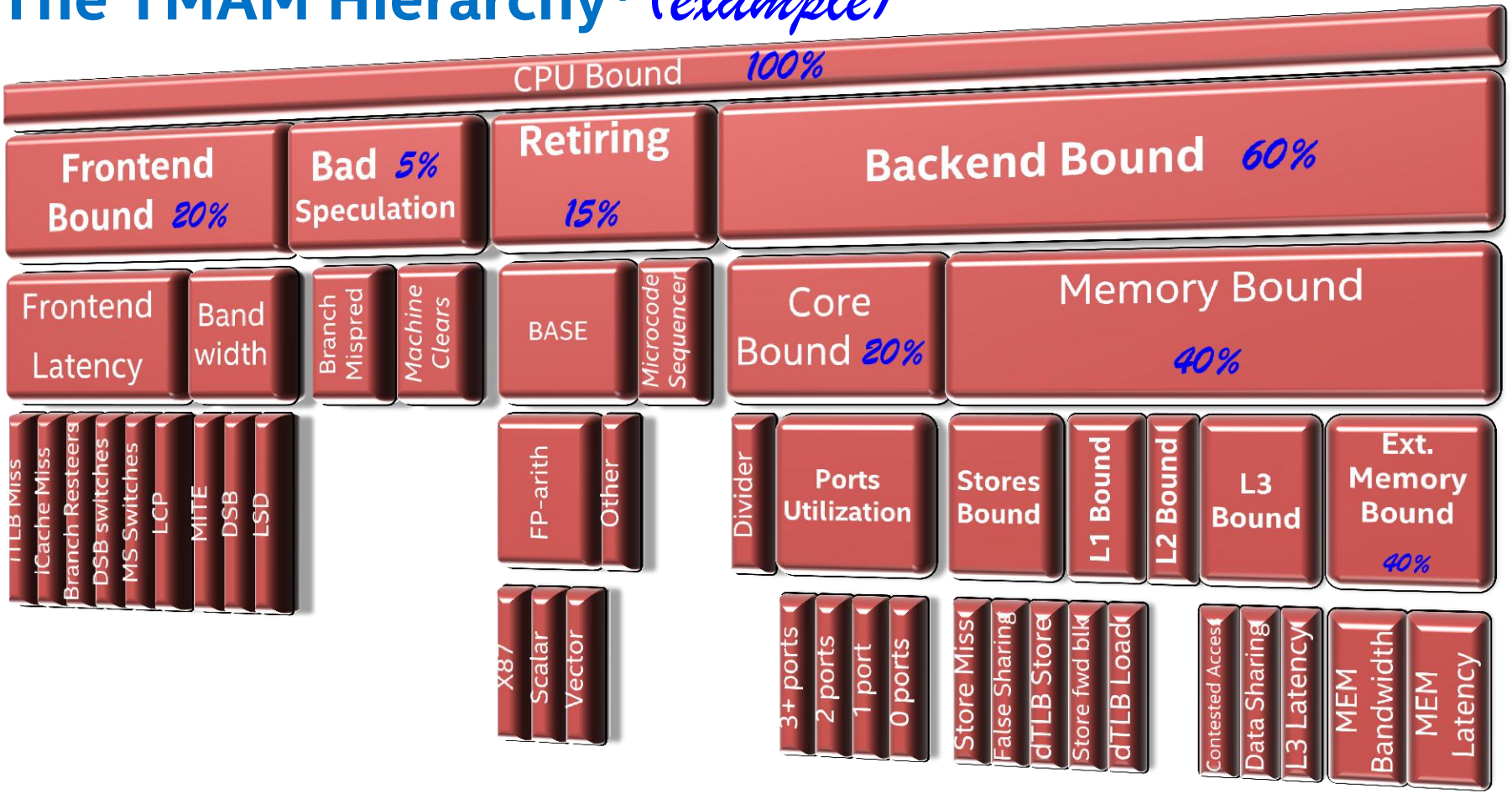
The TMAM Hierarchy[1]



[1] A. Yasin, “A Top-Down Method for Performance Analysis and Counters Architecture”, ISPASS 2014

The TMAM Hierarchy[1] *(example)*



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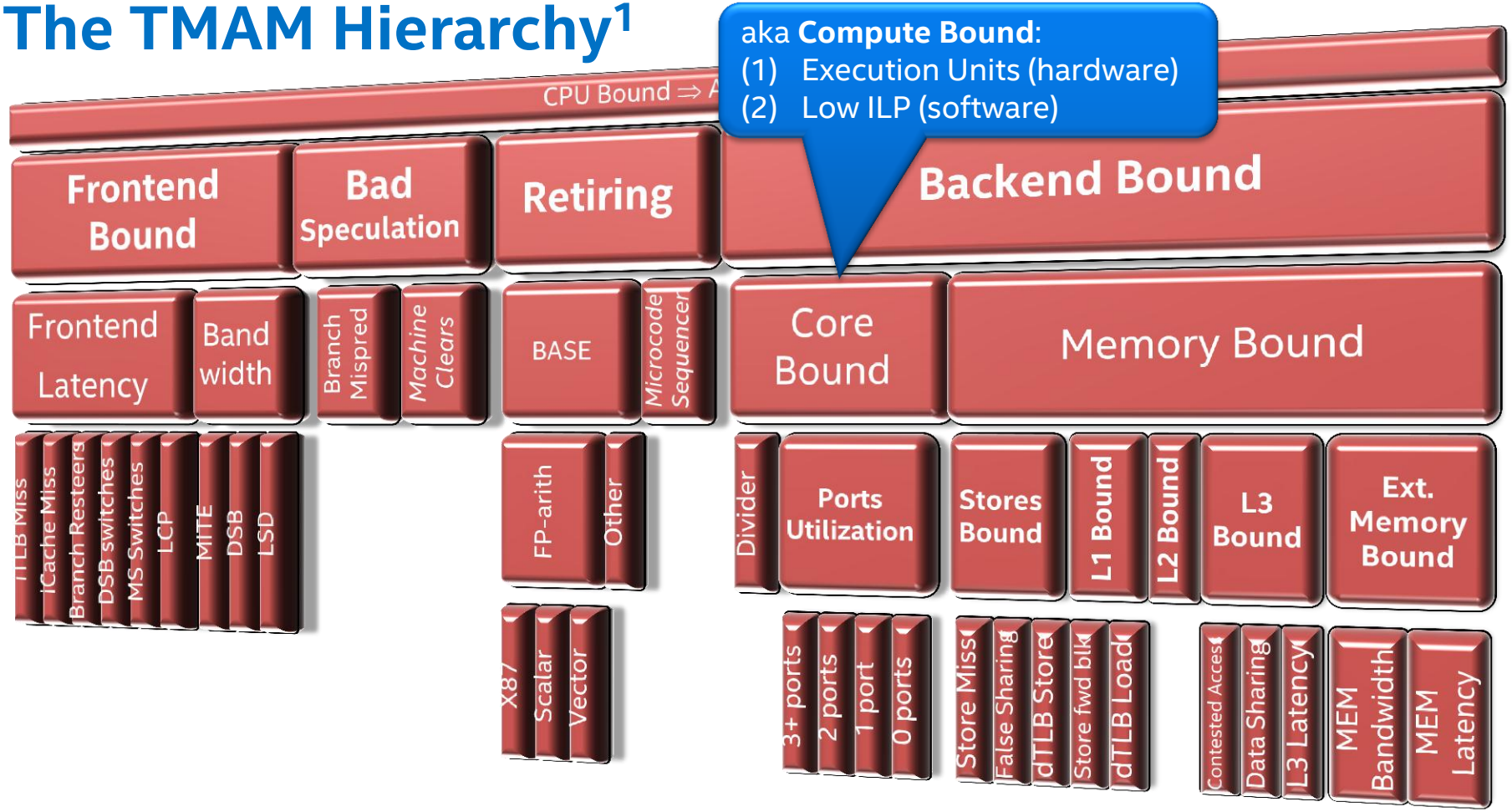
The TMAM Hierarchy[1]



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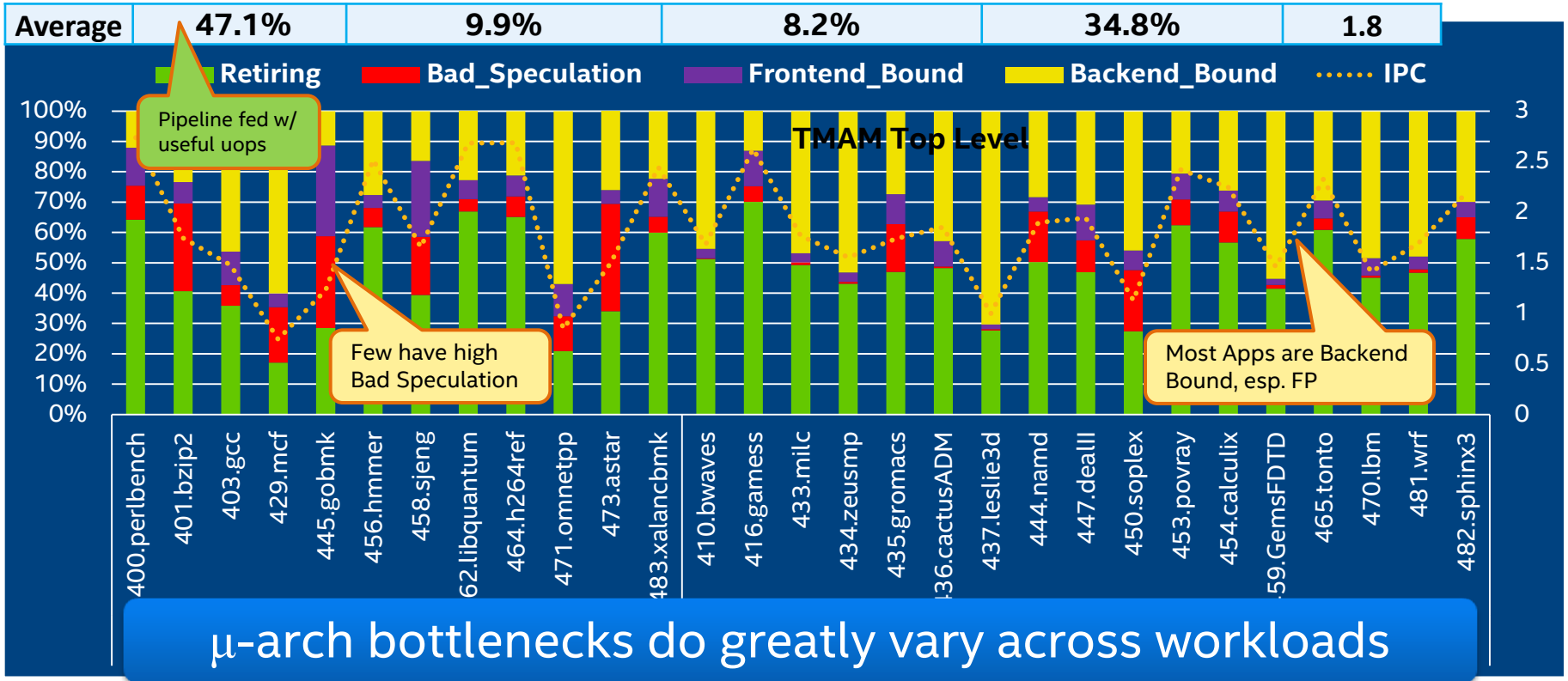
Top Level for SPEC CPU2006



μ-arch bottlenecks do greatly vary across workloads

SPEC CPU2006 v1.2, rate 1-copy, Intel Complier 14 targeting AVX2, Skylake @ 3 GHz

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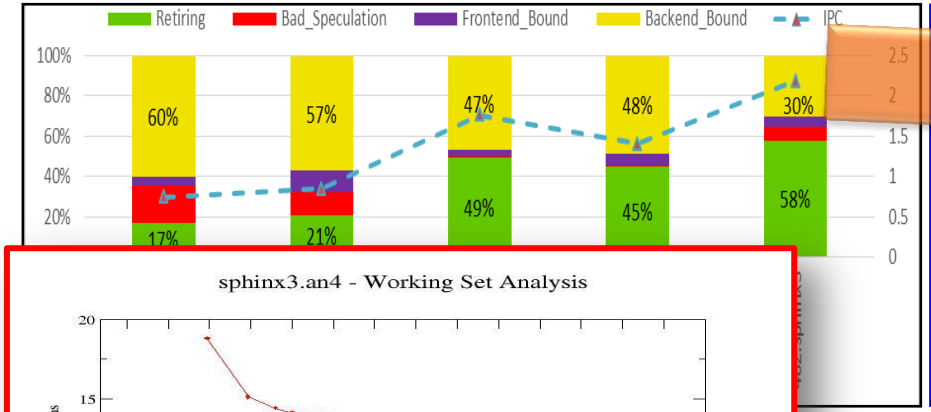
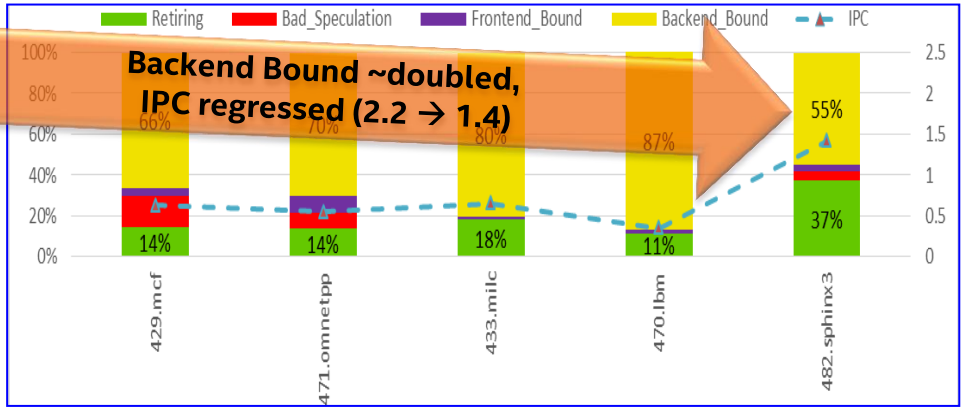
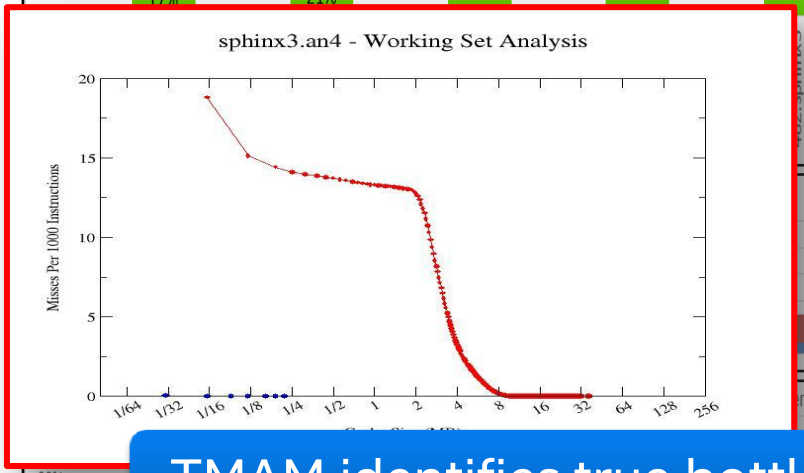
Memory Bound (1-core vs. 4-cores)

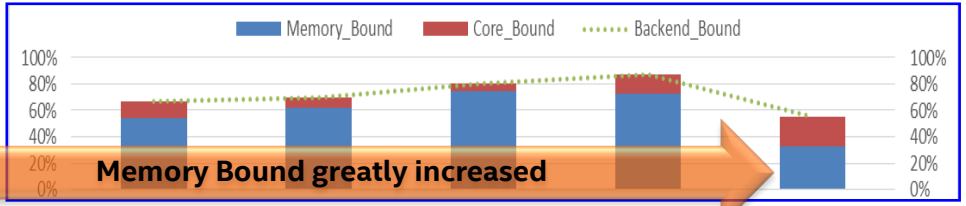
IPC := Instructions Per Cycle





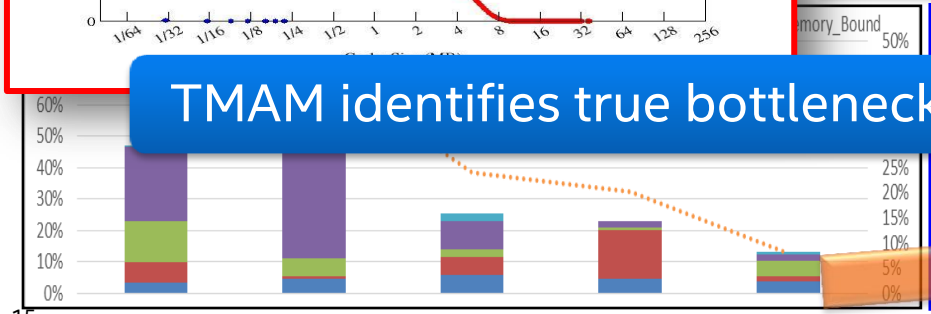
Backend Bound ~doubled, IPC regressed (2.2 → 1.4)

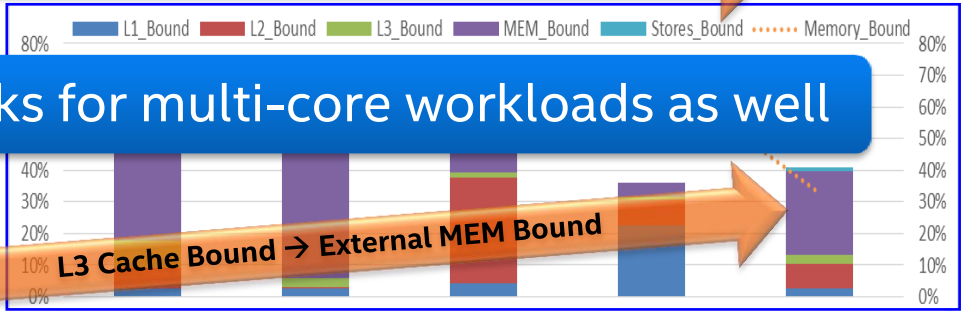




Memory Bound greatly increased

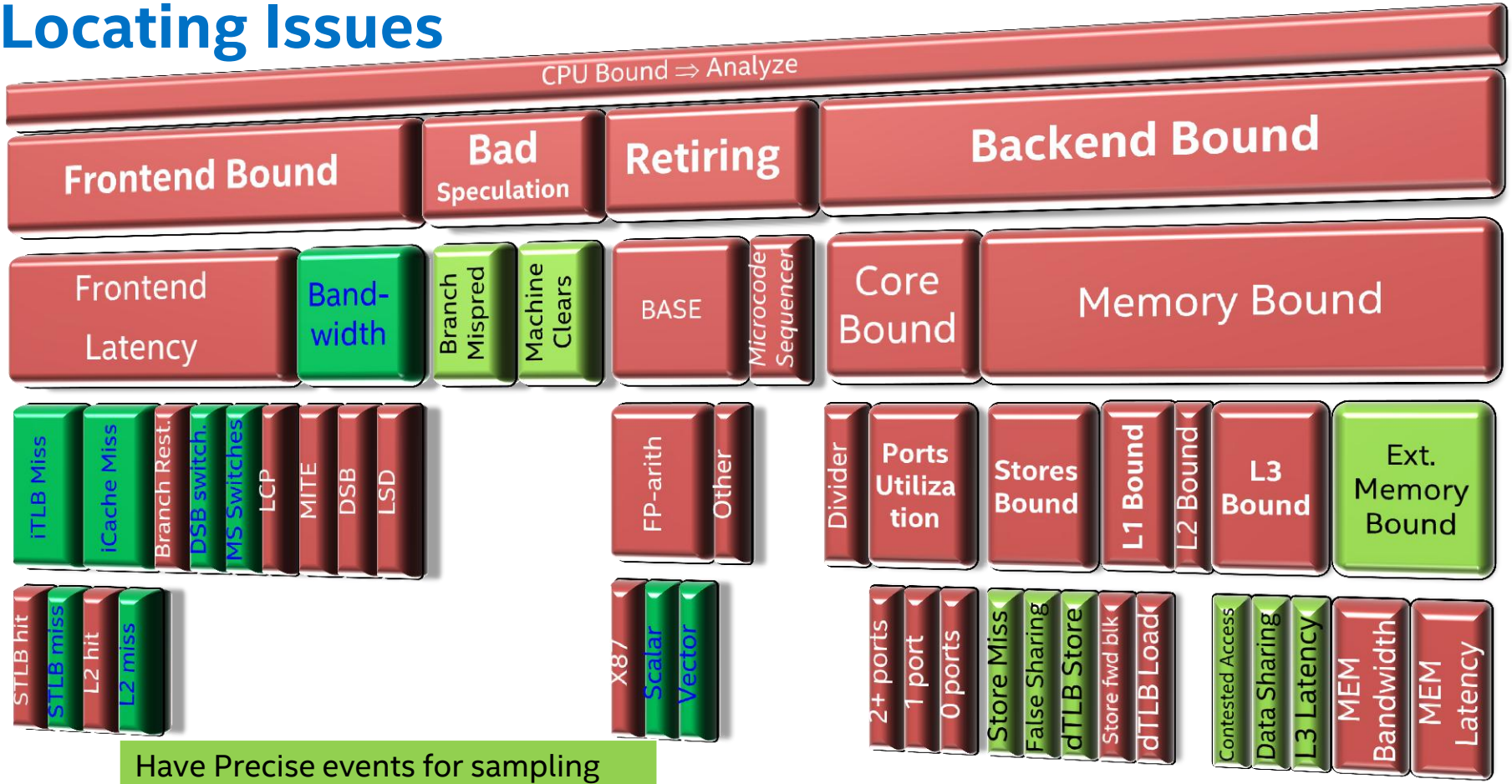
TMAM identifies true bottlenecks for multi-core workloads as well





L3 Cache Bound → External MEM Bound

Locating Issues



Have Precise events for sampling

Precise events added in Skylake

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Move to Demo

Example: Matrix Multiply

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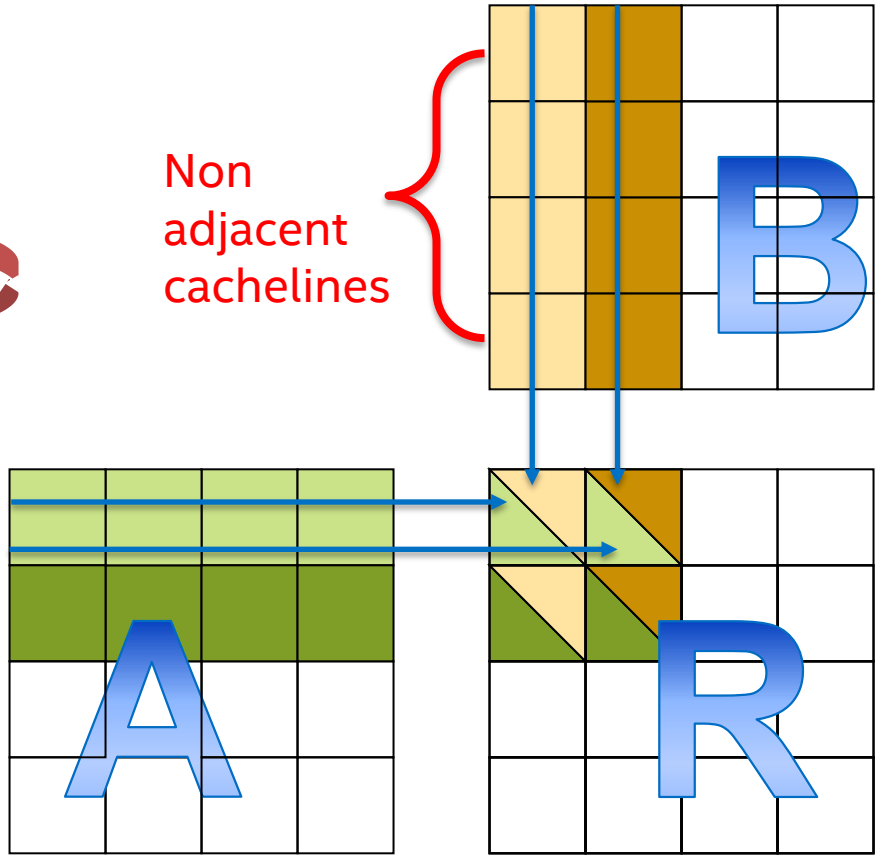
Matrix Multiply insight

```
for (int i=0; i < rows; i++)
  for (int j=0; j < cols; j++)
     for (int k=0 ; k < cols; k++)
         R[i][j] = R[i][j] +
                   A[i][k] * B[k][j];
```



Fix memory access pattern (Loop Interchange optimization)

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DEMO

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VTune's New General Exploration Interface

Front-end Bound

Back-end Bound | Front-End Latency | Front-End Bandwidth

Sort By Front-End Latency

This metric represents a fraction of slots during which CPU was stalled due to front-end latency issues, such as instruction-cache misses, ITLB misses or fetch stalls after a branch misprediction. In such cases, the front-end delivers no uOps.
Formula: ((IDQ_UOPS_NOT_DELIVERED.CYCLES_0_UOPS_DELIV.CORE) / CPU_CLK_UNHALTED.THREAD)

Top-down Tree

Unfilled Pipeline Slots (Stalls)

Function / Call Stack	CPU_CL... THREAD	INST_RETIRED. ANY	CPI Rate	Retiring	Bad Speculation	Front-end Bound
grid_intersect	7,982,000,000	10,640,000,000	0.750	0.316	0.141	0.129
sphere_intersect	7,676,000,000	10,258,000,000	0.748	0.347	0.132	0.034
grid_bounds_intersect	1,192,000,000	826,000,000	1.443	0.176	0.157	0.080
GdipCreateSolidFill	692,000,000	548,000,000	1.263	0.307	0.000	0.087
[TBB Scheduler Internals]	280,000,000	72,000,000	3.889	0.268	0.000	0.705
pos2grid	238,000,000	224,000,000	1.063	0.221	0.200	0.053
[rdpdd.dll]	236,000,000	514,000,000	0.459	0.456	0.117	0.477
tri_intersect	198,000,000	234,000,000	0.846	0.341	0.013	0.038
shader				0.227	0.000	0.028
Raypnt				0.543		0.091
intersect_o				0.116		0.233
VNorm				0.250		0.000
KeSynchron				0.169		0.473

Bad Speculation

Branch Mispredict | Machine Clears

Hover to see Metric description + formula of PMU events, or click arrow to expand a column to see a breakdown of issues pertaining to that category

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Analysis is made easier -- “Follow the Pink”

General Exploration - Sandy Bridge / Ivy Bridge 0 General Exploration viewpoint (chang XE 2013

Analysis Target | Analysis Type | Summary | Bottom-up | Top-down Tree | Tasks and Frames

Grouping: Function / Call Stack

Function / Call Stack	Unfilled Pipeline Slots (Stalls)									
	Back-end Bound									Front-end Bound
	Memory Bound								Core Bound	
	L1 Bound				L2 Bound	L3 Bound	DRAM Bound	Store Bound		
	Loads Blocked by Store Forwarding	Split Loads	4K Aliasing	DTLB Overh ...						
[Loop at line 201 in compute_bodies]					0.000	0.000	0.000	0.000	0.107	0.016
[Loop at line 234 in compute_bodies]					0.000	0.001	0.000	0.004	0.140	0.079
[Loop at line 160 in compute_bodies]					0.000	0.000	0.000	0.000	0.000	0.020
CIsqrt					0.000	0.000	0.000	0.000	0.065	0.033
math_exit					0.002	0.000	0.000	0.000	0.027	0.042
Selected 1 row(s):				0.001	0.000	0.000	0.000	0.000	0.107	0.016

Expand each column marked pink until you come to the actual issue

Here is the problem, read tooltip to learn more

Loads are blocked during store forwarding for a significant proportion of cycles. Use source/assembly view to identify the blocked loads, then identify the problematically-forwarded stores, which will typically be within the ten dynamic instructions prior to the load. If the forwarding store is smaller than the load, change the store to be the same size as the load.

Threshold: ((((13 * LD_BLOCKS.STORE_FORWARD) / CPU_CLK_UNHALTED.THREAD) > 0.05) * (CPU_CLK_UNHALTED.THREAD / > 0.05))

Pick the actual hardware event from the formula (**LD_BLOCKS.STORE_FORWARD – typically counts bigger-size loads blocked by smaller stores to the same address**) for further detailed analysis

Ack: Stas Bratanov

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Un-tuned

General Exploration General Exploration viewpoint (change)

Analysis Target | Analysis Type | Collection Log | Summary | Bottom-up | Top-down Tree | Tasks and Frames

Grouping: Function / Call Stack

Function / Call Stack	Hardware Event C...	Hardware Event...		Filled Pipeline Slots		Unfilled Pipeline Slots (Stalls)	
	CPU_CLK_U... THREAD	INST_RETIRED. ANY	CPI Rate	Retiring	Bad Speculation	Back-end Bound	Front-end Bound
multiply1	488,292,732,438	43,100,064,650	11.329	0.022	0.001	0.974	0.003
KeWaitForMultipleObjects	86,000,129	14,000,021	6.143	0.081	0.244	0.430	0.244
KeSetTimer	86,000,129	6,000,009	14.333	0.000	0.000	0.919	0.081

General Exploration General Exploration viewpoint (change)

Analysis Target | Analysis Type | Collection Log | Summary | Bottom-up | Top-dow | Tasks and Frames | mult

Grouping: Function / Call Stack

Function / Call Stack	Filled Pipeline Slots		Unfilled Pi ts (Stalls)						
			Back-end Bou						
			Memory Bound						
	Retiring	Bad Speculation	L1 Bound	L2 Bound	L3 Bound	DRAM Bound	Store Bound	Core Bound	Front-end Bound
multiply1	0.022	0.001	0.070	0.023	0.064	0.745	0.036	0.022	0.003
KeWaitForMultipleObjects	0.081	0.244	0.000	0.326	0.000	0.000	0.000	0.000	0.244

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Loop Interchange

General Exploration — General Exploration viewpoint (change)

Analysis Target | Analysis Type | Collection Log | Summary | Bottom-up | Top-down Tree | T

Grouping: Function / Call Stack

Function / Call Stack	Hardware Ev...	Hardware Ev...		Filled Pipeline Slots		Unfilled Pipeline Slots (Stalls)	
	CPU_CL... THREAD	INST_RETIRE... ANY	CPI Rate	Retiring	Bad Speculation	Back-end Bound	Front-end Bound
multiply2	43,980,065,970	51,604,077,406	0.852	0.353	0.001	0.573	0.073
KeSetTimer	24,000,036	0	0.000	0.000	0.000	[illegible]	0.000
init_arr	20,000,030	16,000,024	1.250	0.000	0.000	[illegible]	0.000
KeSynchronizeExecution	18,000,027	0	0.000	0.389	0.000	[illegible]	0.000
ExReleaseRundownProte	14,000,021	6,000,009	2.333	0.000	0.000	[illegible]	0.000

General Exploration — General Exploration viewpoint (change)

Analysis Target | Analysis Type | Collection Log | Summary | Bottom-up | Top-down Tree | [illegible] and Frames

Grouping: Function / Call Stack

Function / Call Stack	Unfilled Pipeline Slots (									
	Back-end Bound									
	Memory Bound						Core Bound			
							Port Utilization			
	L1 Bo.	L2 Bou..	L3 Bo.	DRA. Bou..	St. Bo.	DIV Active	Cycles of 0 Ports Utilized	Cycles of 1 Port Utilized	Cycles of 2 Ports Utilized	Cycles of 3+ Ports Utilize
multiply2	0.060	0.000	0.000	0.066	0.137	0.000	0.133	0.353	0.324	0.20
KeSetTimer	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.00
init_arr	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.00
KeSynchronizeExecution	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.00
ExReleaseRundownProte	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.00
Selected 1 row(s):	0.060	0.000	0.000	0.066	0.137	0.000	0.133	0.353	0.324	0.20

Vectorization

Function / Call Stack	CPU_CL... THREAD	INST_RETIRE... ANY	CPI Rate	Retiring	Bad Speculation	Back-end Bound	Front-end Bound
multiply2	43,980,065,970	51,604,077,406	0.852	0.353	0.001	0.573	0.073
KeSetTimer	[illegible]00,036	0	0.000	0.000	0.000	1.000	0.000
init_arr	[illegible],030	16,000,024	1.250	0.000	0.000	1.000	0.000
KeSynchronizeExecution	[illegible]027	0	0.000	0.389	0.000	1.000	0.000
ExReleaseRundownProte	[illegible]000,021	6,000,009	2.333	0.000	0.000	1.000	0.000
Selected 1 row(s):	[illegible]065,970	51,604,077,406	0.852	0.353	0.001	0.573	0.073

General Exp

Analysis Tar

Grouping: Function /

Function / Call Stack	Hardware Eve...	[illegible]are Eve...		Filled Pipeline Slots		Unfilled Pipeline Slots (Stalls)						
						Back-end Bound						
						Memory Bound						
	CPU_CLK... THREAD	[illegible]T_RETIRED. ANY	CPI Rate	Retiring	Bad Speculation	L1 Bo.	L2 Bou..	L3 Bo.	DRAM Bound	Store Bound	Core Bound	Front-end Bound
multiply2	30,810,046,215	24,828,037,242	1.241	0.225	0.001	0.042	0.047	0.010	0.213	0.167	0.172	0.020
ExfAcquirePushLockExclusive	32,000,048	16,000,024	2.000	0.000	0.656	1.000	0.000	0.000	0.000	0.000	0.750	0.000
KeSynchronizeExecution	30,000,045	4,000,006	7.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
init_arr	28,000,042	24,000,036	1.167	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
KeStackAttachProcess	16,000,024	6,000,009	2.667	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Selected 1 row(s):	30,810,046,215	24,828,037,242	1.241	0.225	0.001	0.042	0.047	0.010	0.213	0.167	0.172	0.020

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Vectorization: IvyBridge vs. Skylake

General Exploration General Exploration viewpoint (change)

Analysis Target | Analysis Type | Collection Log | Summary | Bottom-up | Top-down Tree | Tasks and Frames | multiply.c

Grouping: Function / Call Stack

Function / Call Stack	Hardware Eve... CPU_CLK... THREAD	Hardware Eve... INST_RETIRED. ANY	CPI Rate	Filled Pipeline Slots: Retiring	Filled Pipeline Slots: Bad Speculation	Unfilled Pipeline Slots (Stalls): Back-end Bound: Memory Bound: L1 Bo.	L2 Bou..	L3 Bo.	DRAM Bound	Store Bound	Core Bound	Front-end Bound
multiply2	30,810,046,215	24,828,037,242	1.241	0.225	0.001	0.042	0.047	0.010	0.213	0.167	0.172	0.020
ExfAcquirePushLockExclusive	32,000,048	16,000,024	2.000	0.000	0.656	1.000	0.000	0.000	0.000	0.000	0.750	0.000
KeSynchronizeExecution	30,000,045	4,000,006	7.50	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

General Exploration General Exploration viewpoint (change) Intel VTune Amplifier XE 2016

Collection Log | Analysis Target | Analysis Type | Summary | Bottom-up | PMU Events | Platform

Grouping: Function / Call Stack

Function / Call Stack	Clockticks	Instructions Retired	CPI Rate	Front-End Bound	Bad Speculation	Back-End Bound: Memory Bound	Back-End Bound: Core Bound	Retiring
multiply3	15,678,023,517	24,822,037,233	0.632	0.001	0.001	0.318	0.301	0.379
init_arr	34,000,051	32,000,048	1.063	0.000	0.000	0.000	1.000	0.000

 Intel® Microarchitecture, Code Name Skylake. Intel® Microarchitecture, Code Name Ivy Bridge

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FMA (Fused Multiply Add)

General Exploration General Exploration viewpoint (change) Intel VTune Amplifier XE 2016

Collection Log | Analysis Target | Analysis Type | Summary | Bottom-up | PMU Events | Platform

Grouping: Function / Call Stack

Function / Call Stack	Clockticks	Instructions Retired	CPI Rate	Front-End Bound	Bad Speculation	Back-End Bound: Memory Bound	Back-End Bound: Core Bound	Retiring
multiply3	15,678,023,517	24,822,037,233	0.632	0.001	0.001	0.318	0.301	0.379

Function / Call Stack	Clockticks	Instructions Retired	CPI Rate	Front-End Bound	Bad Speculation	Back-End Bound: Memory Bound	Back-End Bound: Core Bound	Retiring
multiply4	10,696,016,044	10,332,015,498	1.035	0.015	0.000	0.511	0.266	0.208
func@0x14015ae00	40,000,060	4,000,006	10.000	0.000	0.000	0.000	1.000	0.000
init_arr	16,000,024	34,000,051	0.471	0.000	0.000	0.000	1.000	0.000
func@0x1400c0900	12,000,018	2,000,003	6.000	0.000	0.000	0.000	1.000	0.000

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Matrix Multiply Summary

A function is *iteratively* analyzed

1. Big matrices in memory
   - `multiply1` is **Ext. Memory Bound**
2. Loop Interchange
   - `multiply2` becomes **Core Bound** due to execution ports high utilization
3. Vectorization
   - `multiply3` reduces ports pressure but workload is still **Core Bound** and **Memory Bound**
4. FMA – Fused Multiply-Add
   - `multiply4` further reduces instruction count with better latency/throughput in Skylake. Workload back to **Memory Bound**

step	Optimization	Time [s]	speedup
1	Textbook (baseline)	41.3	1.0x
2	Loop Interchange	5.1	8.1x
3	+Vectorization	4.1	10.1x
4	+FMA	2.8	14.7x

Performance bottlenecks vary as the code is optimized

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Agenda

- Performance Analysis & Challenges
- Top-down Microarchitecture Analysis Method
- Demo using Intel® VTune™ Amplifier XE
- Skylake Performance Monitoring Unit Enhancements
- Summary and Q/A

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Intel® Core™ Processor PMU Refresher

- Architectural PMU (Performance Monitoring Unit)
  - 3 fixed counters
    - Instructions Retired, Core- and Reference-Unhalted-Cycles
  - 4 general-purpose counters
    - Expands to 8 when SMT is off
  - Global Status and Controls
- Rich list of performance events
  - Most useful events are model-specific
  - A subset of 7 events are architected
- Advanced mechanisms
  - PEBS: Precisely tags performance events & profiling data into software location
  - LBR: Non-intrusive branch recording facility

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Skylake PMU Overview

- Architectural PerfMon v4
- Enriched event-list
  - **Enhancing TMAM** with more top-down oriented events
  - Precise Front-end Events
- PEBS Enhancements
  - Included TimeStamp Counter (TSC)
  - Fixed multi-counter PEBS x-talk issue
- LBR Enhancements
- Intel Processor Trace v2 facilitates responsiveness analysis

	Feature	Haswell	Skylake
Arch PerfMon	Version	3	4
	VMM/OS can virtualize Global Status	No	Yes
	Robust Interface for freeze on PMI		Better/faster
	In-use register		Architectural
	Secure mode	n/a	Yes (w/ SGX)
PEBS	Frontend Events	No	Yes
	TSC in record	No	Yes
	Multi counter indication	Inaccurate	Accurate
LBR	Timing info	No	Yes
	# entries	16	32
	# MSRs / entry	2	3

PEBS – Precise Events Based Sampling. LBR – Last Branch Records

Intel® Microarchitecture, Code Name Skylake; Intel® Microarchitecture, Code Name Haswell

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Precise Front-end Events

- Precise Event-Based Sampling (PEBS)
  - Reports precise information like Instruction address of cache misses
- Skylake expands PEBS to support Front-end events
  - These are the hardest to locate in code since they are the farthest from retirement
  - Adding at-retirement version of five key miss events in addition to a generic event

Event	Sub-Event	Description
FRONTEND_RETIRED	L1I_MISS	Retired Instructions following an i-cache L1 miss
	L2_MISS	Retired Instructions following an i-cache L2 miss
	ITLB_MISS	Retired Instructions following an iTLB miss (1st level)
	STLB_MISS	Retired Instructions following an iTLB & STLB miss (2nd level)
	DSB_MISS	Retired Instructions following a DSB (uop cache) miss
	IDQ_READ_BUBBLES	Retired Instructions following a “certain” IDQ read bubbles pattern (configurable latency/BW conditions). Example: periods with uop delivery starvation ≥x cycles

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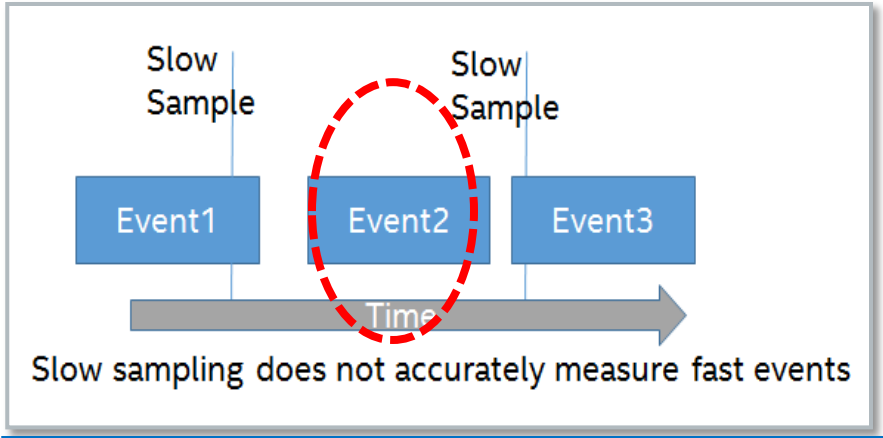
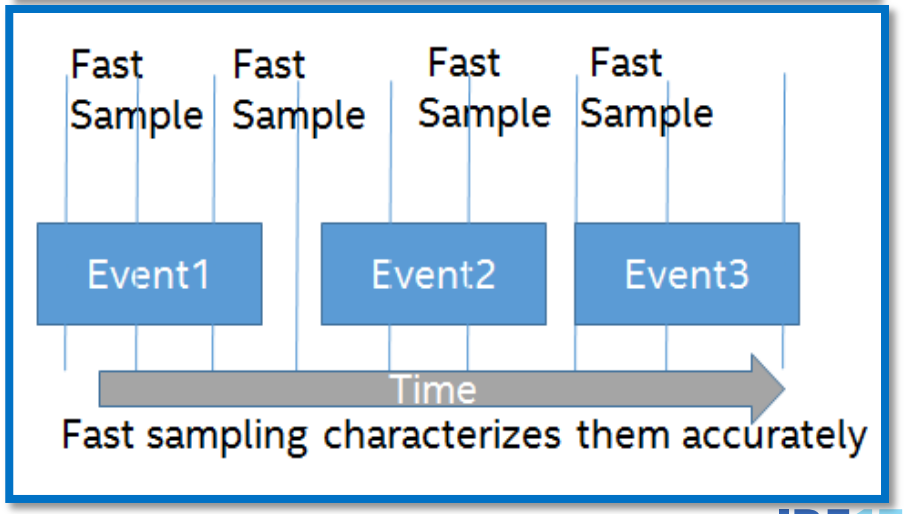
PEBS Enhancements

- Skylake PEBS features improve interrupt less sampling
  - Timestamp (TSC) with each record
  - Applicable Counters bit-field
- Newly Enabled usages
  - Low overhead profiling
    - Interrupt less sampling allows fast sampling at reasonable overhead
  - Profiled regions with masked interrupts
  - Multi record PEBS buffering for system-wide profiling





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Accurate basic-block timing

- LBRs - Last Branch Records
  - A buffer of recently taken branches
- Skylake Enhancements
  - **Timed LBRs**
    - **Accurate # of cycles between taken branches**
    - **Key enabler for new usages – examples:**
      - Time function calls; loop iterations,
      - Min/max/avg time per basic-block,
      - Potential speedup when using a perf simulator ➔
  - 2x Buffer size
    - Skylake doubled number of entries from 16 up to 32

Today

Address	Source Line	Assembly	Hardware Event Count CPU_CLK_UNHALTED...
0x401005		**Block 1:**	
0x401005		jmp 0x411950 <Block 2>	7,404,306
0x411950		**Block 2:**	
0x411950	1	push ebp	10,266,120
0x411951	1	mov ebp, esp	177,420
0x411953	1	push ecx	
0x411954	3	mov eax, dword ptr [ebp+0x8]	10,058,619
0x411957	3	add eax, dword ptr [ebp+0xc]	5,094,074
0x41195a	3	mov dword ptr [ebp-0x4], eax	5,500,459
0x41195d	4	mov esp, ebp	14,726,529
0x41195f	4	pop ebp	2,468,698
0x411960	4	ret	131,662

Potential usage

Function / Call Stack	Hardware Event Count CPU_CLK_UNHALTED....	CPI Rate	Ideal CPI	Potential Speedup
main	166,948,980	0.427	**0.262**	**37%**
multiply	61,921,921	0.421	**0.400**	**5%**
add	55,827,887	0.408	**0.286**	**30%**

Intel® Microarchitecture, Code Name Skylake;

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Summary

- TMAM: Top-down µarch Analysis Method
  - Simplifies Performance Analysis using a structured hierarchy
    - Eliminates guess work
    - Reduces microarchitecture high learning curve
  - Forward compatibility, consistency across IA processors
- Skylake PMU adds key capabilities
  - Precise Front-end events
  - Better Sampling
  - Fine granular timing

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Additional Sources of Information

- A PDF of this presentation is available from our Technical Session Catalog: www.intel.com/idfsessionsSF. This URL is also printed on the top of Session Agenda Pages in the Pocket Guide.
- Technical info
  - Intel® Architecture Optimization Reference Manual, Top-Down Analysis Method chapter
  - A Top-Down Method for Performance Analysis and Counters Architecture, Ahmad Yasin. *In IEEE International Symposium on Performance Analysis of Systems and Software, ISPASS 2014.* [paper] [slides]
- More web based info:
  - https://software.intel.com/en-us/intel-vtune-amplifier-xe/try-buy (a free 30 day evaluation VTune)
  - https://github.com/andikleen/pmu-tools/wiki/toplev-manual (Linux perf wrapper by Andi Kleen)
  - Native Linux perf support for TMAM-Level1 is coming soon.
  - Developers: check out TMAM-metric files: https://download.01.org/perfmon/

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Other Technical Sessions

Session ID	Title	Day	Time	Room
✓ ARCS001	Intel® Architecture, Code Name Skylake Deep Dive: A New Architecture to Manage Power Performance and Energy Efficiency	Tues	1:15	2006
✓ ARCS002	Software Optimizations Become Simple with Top-Down Analysis Methodology on Intel® Microarchitecture, Code Name Skylake	Tues	2:30	2006
ARCS003	Intel® Architecture Code Name Skylake Deep Dive: Hardware-Based Security for Windows® 10	Tues	4:00	2006
Special	Zoom-in on Your Code with Intel® Processor Trace and Supporting Tools	Tues	5:30	Showcase Networking Plaza

✓ = DONE

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Completing an Online Session Evaluation by 10am Tomorrow Automatically Enters You in a Drawing to Win!

You will receive an email with a link to the online evaluation prior to the end of this session.

Day 1 Prize

Intel® Compute Stick (20)

Day 2 Prize

Microsoft* Surface* 3 (6)

Day 3 Prize

Dell Venue 10 7000 Series (4)

Winners will be notified by email

Copies of the complete sweepstakes rules are available at the Info Desk

Q&A

ahmad.yasin@intel.com

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Backup

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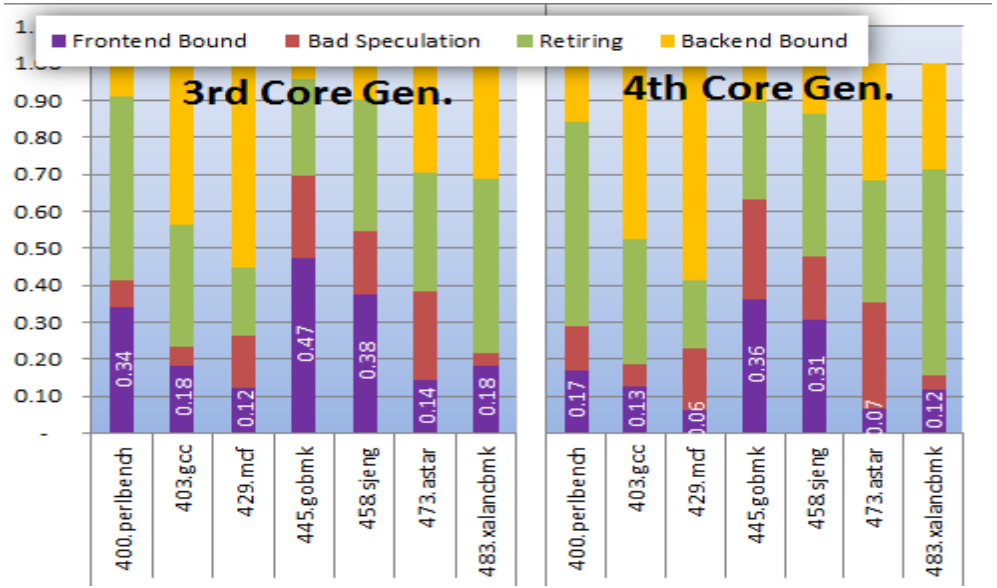
Microarchitecture Comparison

- Haswell (4th generation Intel® Core™ processor) has improved front-end
  - Speculative i-TLB and i-cache accesses with better timing to improve the benefits of prefetching
- Benefiting benchmarks clearly show reduction in Frontend Bound



Using TMAM, forward compatibility is assured on Intel Core

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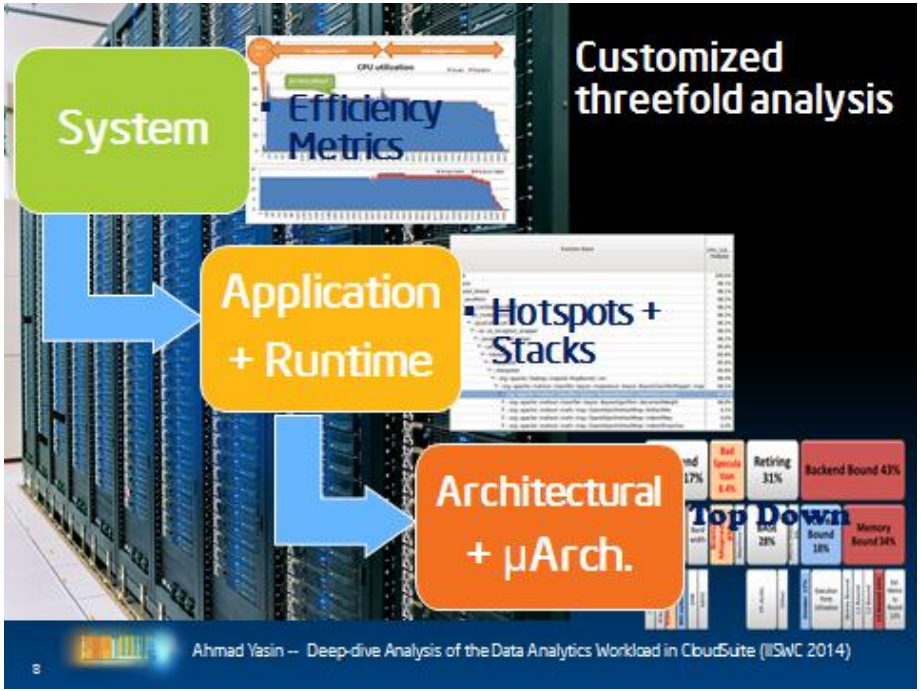
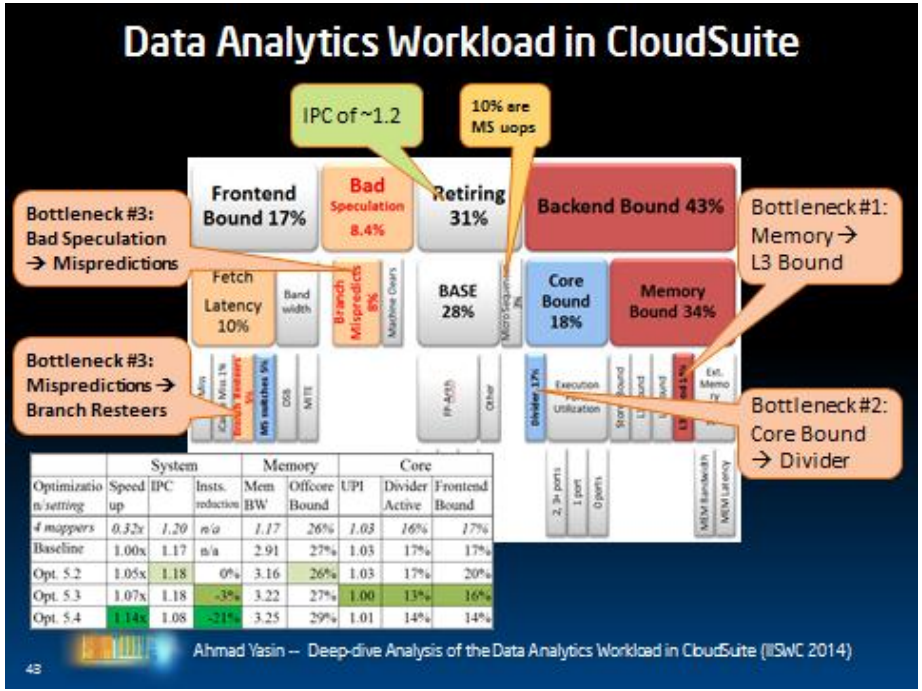
Sample Workload Optimization

Deep-dive Analysis of the Data Analytics Workload in CloudSuite - Ahmad Yasin, Yosi Ben-Asher, Avi Mendelson. *In IEEE International Symposium on Workload Characterization, IISWC 2014.* [paper] [slides]





	System			Memory		Core		
Optimization setting	Speed up	IPC	Insts. reduction	Mem BW	Offcore Bound	UPI	Divider Active	Frontend Bound
4 mappers	*0.32x*	*1.20*	*n/a*	*1.17*	*26%*	*1.03*	*16%*	*17%*
Baseline	1.00x	1.17	n/a	2.91	27%	1.03	17%	17%
Opt. 5.2	1.05x	1.18	0%	3.16	26%	1.03	17%	20%
Opt. 5.3	1.07x	1.18	-3%	3.22	27%	1.00	13%	16%
Opt. 5.4	1.14x	1.08	-21%	3.25	29%	1.01	14%	14%

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Datacenter Profiling

- Profiling a Warehouse-Scale Computer - S. Kanev, J. P. Darago, K. Hazelwood, P. Ranganathan, T. Moseley, G. Wei and D. Brooks, in International Symposium on Computer Architecture (ISCA), June 2015.

- First to *profile* a production datacenter
  - Mixture of μ-arch bottlenecks
    - Stalled on data most often
    - Heavy pressure on i-cache
    - Compute in bursts
    - Low memory BW utilization

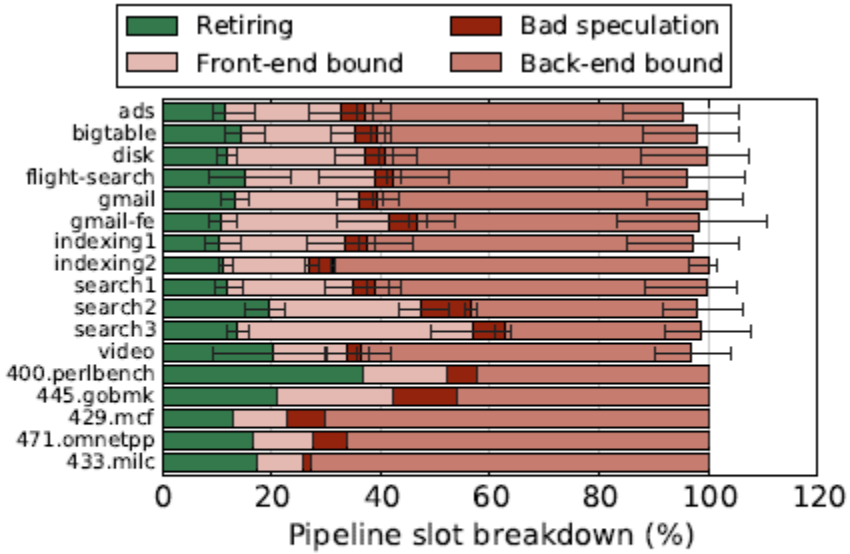

Figure 4: Top-level bottleneck breakdown. SPEC CPU2006 benchmarks do not exhibit the combination of low retirement rates and high front-end boundedness of WSC ones.

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Arch PerfMon Version 4

- Better support to *virtualize* & *share* the Architectural PMU
  - IA32_PERF_GLOBAL_STATUS (0x38E) had per-counter overflow status bit
  - New IA32_PERF_GLOBAL_STATUS_SET enables setting bits in 0x38E
    - Enables a VMM to virtualize overflow status bits (only)
  - New Counters Frozen interface
  - New LBRs Frozen interface
  - In addition, IA32_PERF_GLOBAL_INUSE facilitates sharing the PMU among multiple agents (Architectural register at address 0x392)

IA32_PERF_GLOBAL_STATUS (0x38E)

63	62	61	60	59	58	57:35	34	33	32	31:8	7	6	5	4	3	2	1	0
CondChg	BufferOvf	UnCore Ovf	interference*	Counters Frozen	LBRs Frozen	Rsvd	OvrFxCtr 2	OvrFxCtr 1	OvrFxCtr 0	Rsvd	OvrCtr 7	OvrCtr 6	OvrCtr 5	OvrCtr 4	OvrCtr 3	OvrCtr 2	OvrCtr 1	OvrCtr 0
0	0	0	0	0	0	0 (GP)	0	0	0	0 (GP)	0 (wr ´1 #GP if HT on)				0	0	0	0

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